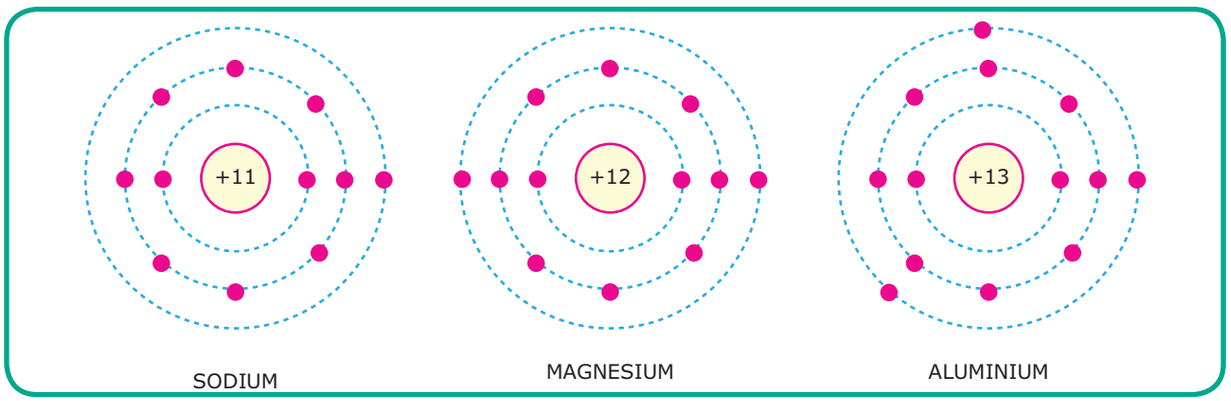


Figure 3.3 Atomic structure of sodium, magnesium and aluminium

When number of valence electron of an atom is more than 4, the material is usually a non-metal and an insulator.

Examples are nitrogen, sulphur and neon which have 5, 6 and 8 valence electrons respectively as shown in Figure3.4.

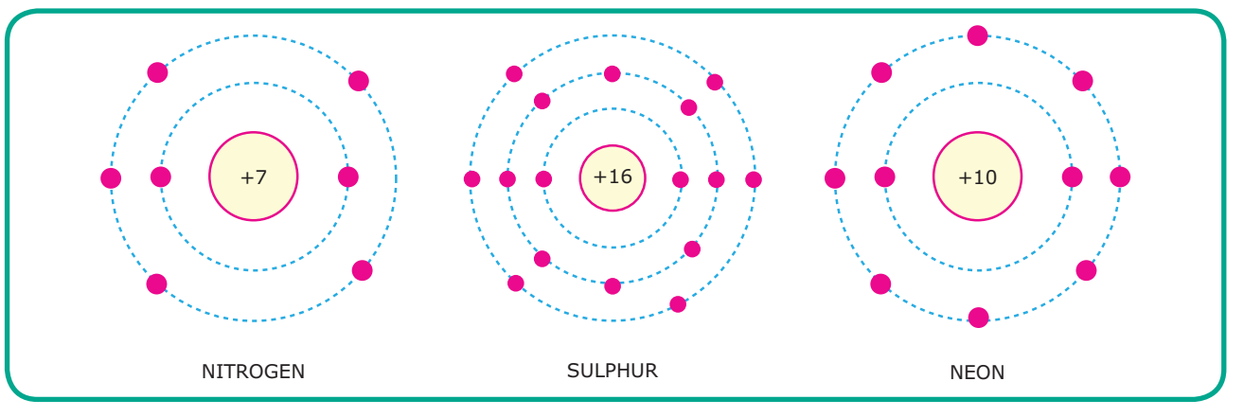


Figure 3.4 Atomic structure of nitrogen, sulphur and neon

iii. Semiconductor:

When the number of valence electrons of an atoms is 4 (i.e., exactly one-half of the maximum of 8 electron), the material has both metal and non-metal properties and is usually a semiconductor. Examples are carbon, silicon and germanium as shown in below Figure3.5.

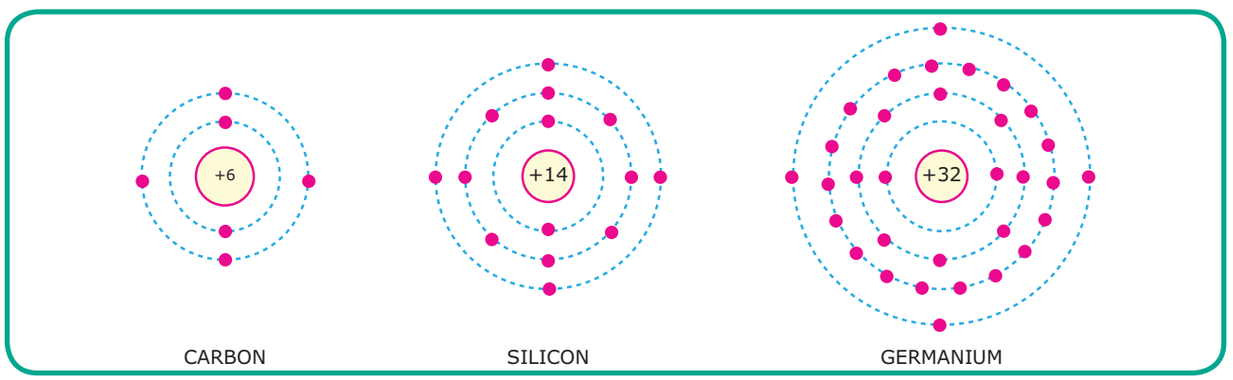


Figure 3.5 Atomic structure of carbon, silicon and germanium

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